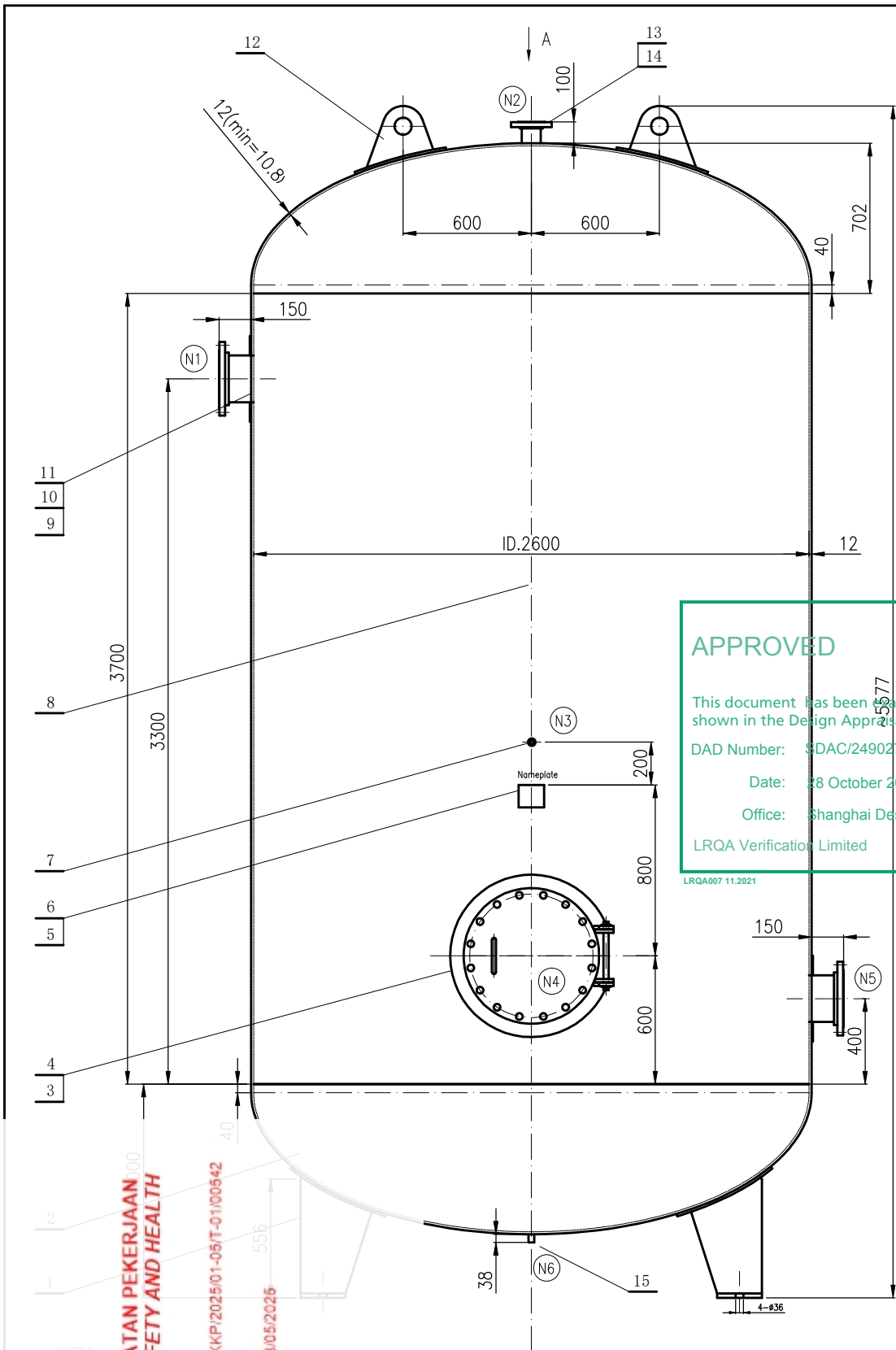




APPROVED

This document has been examined and given the status as shown in the Design Appraisal Document (DAD) number below:

DAD Number: S-DAC/2490271-08

Date: 28 October 2024 Initials: WL

Office: Shanghai Design Appraisal Center

LRQA Verification Limited

A member of the LRQA Group Limited

LRQA007 11.2021

Longitudinal, Circumferential welds

纵环焊缝焊接视图

EXTERNAL外侧

60°±5°

INTERNAL内侧

2±2

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

12

View at A

A向视图

0°

315°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°

0°

45°

90°

135°

180°

225°

270°

315°

360°



JABATAN KESELAMATAN DAN KESIHATAN PEKERJAAN  
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH

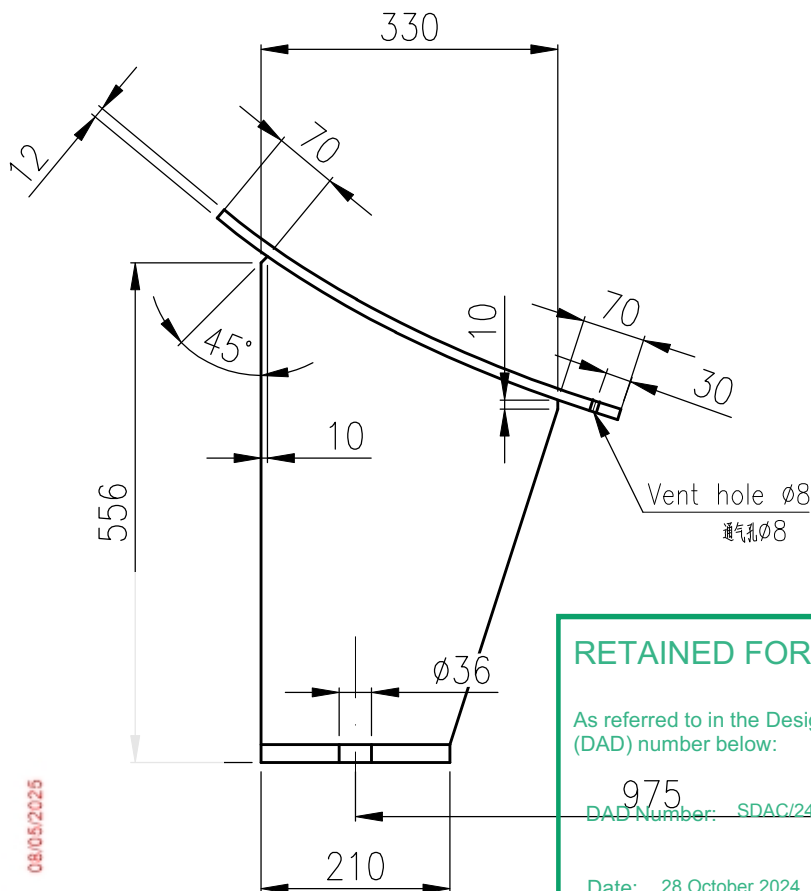
NO. VERIFIKASI: JKKP/2025/01-05/T-01/00542

TARIKH: 08/05/2025



Verifikasi Dokumen  
Document Verification

ES:  
Backing material: SA-516M Gr. 185. Rib and base plate material: SA/GB713 Q345R.  
筋板及底板材料: SA/GB713 Q345R.  
The fields is the same as the thinner plate.  
为准。  
fit the outer wall of the vessel.  
welding of the support, all parts shall be flat  
without warping.  
支座组焊完毕后, 各部件应平整, 不得翘曲。



### RETAINED FOR INFORMATION

As referred to in the Design Appraisal Document (DAD) number below:

DAD Number: 975 SDAC/2490271-08

Date: 28 October 2024

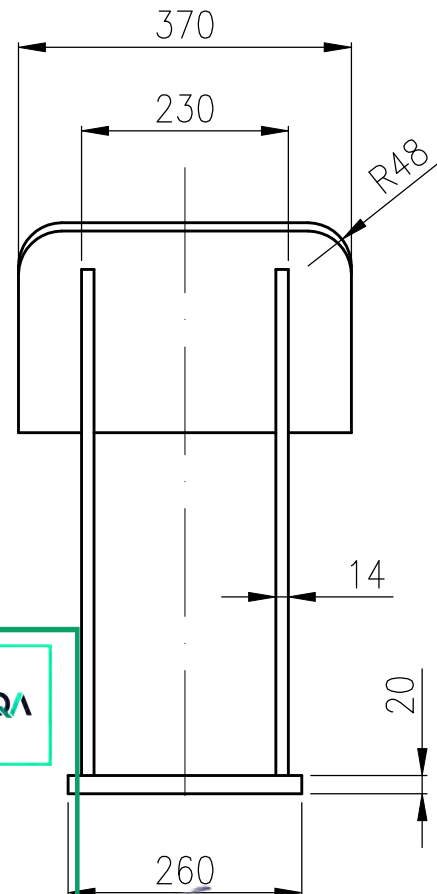
Office: Shanghai Design Appraisal Centre

LRQA Verification Limited

LRQA

Initials: WL

A member of the LRQA Group Limited



|             |         |           |      |                             |        |             |      |                                                         |  |
|-------------|---------|-----------|------|-----------------------------|--------|-------------|------|---------------------------------------------------------|--|
|             |         |           |      | Support<br>支座               |        |             |      | 浙江泰克压力容器有限公司<br>Zhejiang Tank Pressure Vessel Co., Ltd. |  |
| Rev.        | Comment | Signature | Date | 重量<br>Weight                | 67.2kg | 比例<br>Scale | 1:10 |                                                         |  |
| Design by   |         | Date      |      | Drawing NO. :<br>TK24324-01 |        |             |      |                                                         |  |
| Approved by |         | Date      |      | Rev. A                      |        |             |      |                                                         |  |



JABATAN KESELAMATAN DAN KESIHATAN PEKERJAAN  
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH

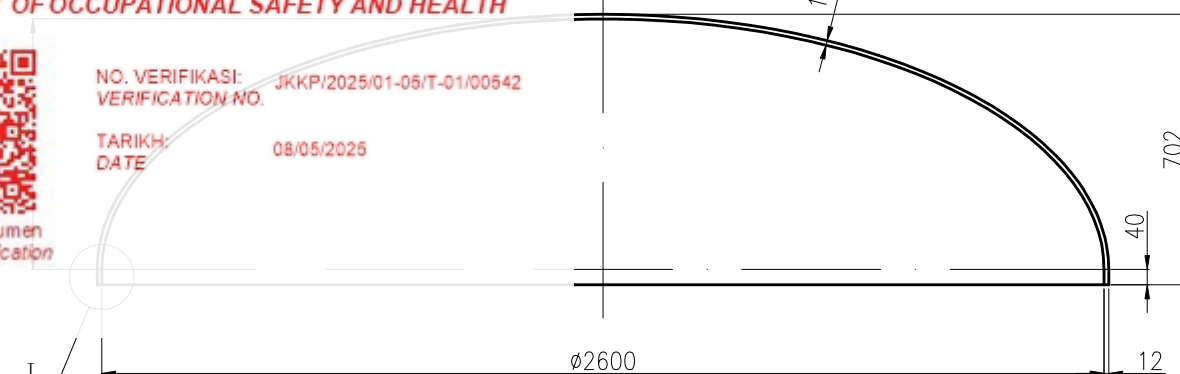
others



Verifikasi Dokumen  
Document Verification

NO. VERIFIKASI: JKKP/2025/01-05/T-01/00542  
VERIFICATION NO.

TARIKH: 08/05/2025  
DATE



APPROVED

LRQA

This document has been examined and given the status as shown in the Design Appraisal Document (DAD) number below:

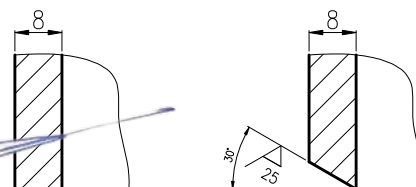
DAD Number: SDAC/2490271-08

Date: 28 October 2024 Initials: WL

Office: Shanghai Design Appraisal Cent

LRQA Verification Limited

A member of the LRQA Group Limited



Detail I (详图 I)

NOTES: LRQA007 11.2021

1. Code 规范: ASME VIII-1, 2023 ED, No stamp;
2. MAWP 最高允许工作压力: 0.84 MPa at 100°C;
3. MDMT 最低金属设计温度: -18°C at 0.84 MPa;
4. Corrosion allowance 腐蚀裕量: 1.0 mm;
5. Material 材料: SA-516M Gr. 485;
6. NDE: Spot RT as per UW-52; 无损检测: 局部RT;
7. Impact test: Exemption (UCS-66); 冲击试验: 豁免 (UCS-66);
8. Heat treatment after forming: Exemption (UCS-56). 成形后热处理: 豁免 (UCS-56);
9. The minimum thickness of ellipsoidal head after forming shall be 10.8 mm.

封头成型后最小厚度为10.8 mm.

10. As per UG-81, the inner surface of a ellipsoidal head shall not deviate outside of the specified shape by more than 1.25%D nor inside the specified shape by more than 5/8% of D. The skirts of heads shall be sufficiently true to round so that the difference between the maximum and minimum inside diameters shall not exceed 1% of the nominal diameter.

根据UG-81, 椭圆形封头的内表面与规定形状(样板)的向外偏差不得大于1.25%D(32.5mm), 且与规定形状的向内偏差不得大于5/8%D(16.25mm)。封头的直边应该有足够的圆度, 使最大及最小内径之差不超过公称直径的1%。

11. Other size requirement refer to GB/T25198-2023.

其他尺寸要求参照GB/T25198-2023。

12. The head groove of each vessel shall be one flat groove and one outer slope according to Detail I.

每台容器的封头坡口按照详图I, 平坡口和外坡口各一件。

|             |               |                 |            |                                      |         |             |      |                                                         |  |
|-------------|---------------|-----------------|------------|--------------------------------------|---------|-------------|------|---------------------------------------------------------|--|
|             |               |                 |            | 2:1 ellipsoidal head<br>2:1椭圆封头      |         |             |      | 浙江泰克压力容器有限公司<br>Zhejiang Tank Pressure Vessel Co., Ltd. |  |
| 版本<br>Rev.  | 注释<br>Comment | 签名<br>Signature | 日期<br>Date |                                      |         |             |      |                                                         |  |
| Design by   |               | Date            |            | 重量<br>Weight                         | 705.5kg | 比例<br>Scale | 1:25 |                                                         |  |
| Approved by |               | Date            |            | Drawing NO.:<br>TK24324-02<br>Rev. A |         |             |      | Customer:                                               |  |

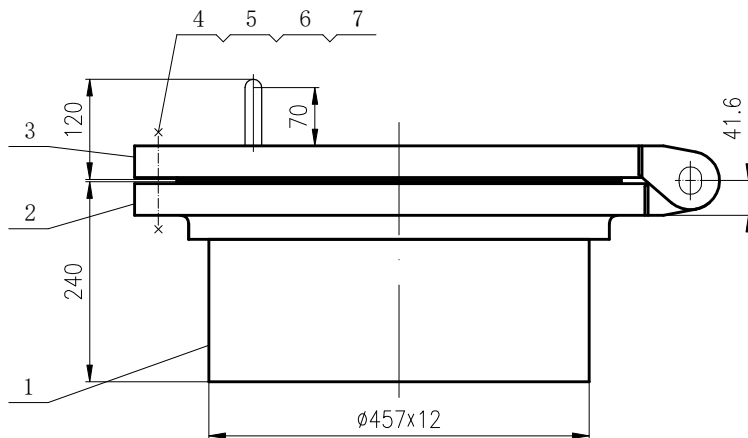


NO. VERIFIKASI: JKKP/2025/01-05/T-01/00542  
VERIFICATION NO.

TARIKH: 08/05/2025  
DATE



Verifikasi Dokumen  
Document Verification



APPROVED

LRQA

This document has been examined and given the status as shown in the Design Appraisal Document (DAD) number below:

DAD Number: SDAC/2490271-08

Date: 28 October 2024 Initials: WL

Office: Shanghai Design Appraisal Cent

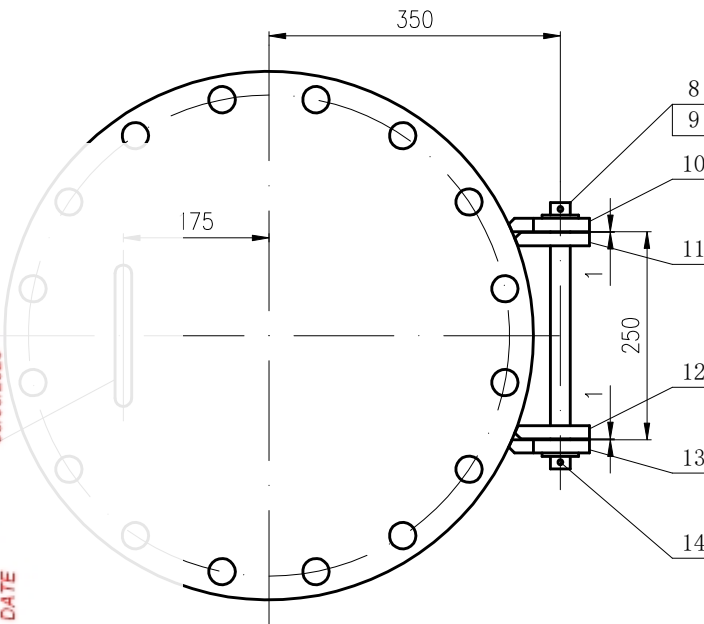
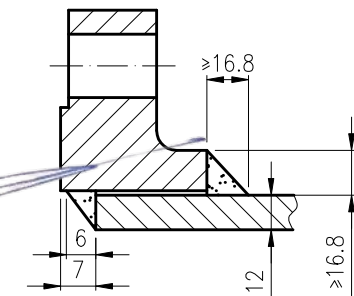
LRQA Verification Limited

A member of the LRQA Group Limited

LRQA007 11.2021

Welding detail for flange  
to pipe

法兰与接管焊接视图



|             |                 |                             |            |                                                  |                              |                   |                |  |
|-------------|-----------------|-----------------------------|------------|--------------------------------------------------|------------------------------|-------------------|----------------|--|
| 15          | TK24324-03-06   | Handle 把手 Ø20               | 1          | SA-105M                                          |                              | 0.7               |                |  |
| 14          | GB/T91-2000     | Pin 销 6.3×30                | 2          | Q215                                             | /                            | /                 |                |  |
| 13          | TK24324-03-05   | Cover lug 盖轴耳 II            | 1          | SA/GB713 Q345R                                   |                              | 0.7               |                |  |
| 12          | TK24324-03-04   | Flange lug 法兰轴耳 II          | 1          | SA/GB713 Q345R                                   |                              | 0.6               |                |  |
| 11          | TK24324-03-03   | Flange lug 法兰轴耳 I           | 1          | SA/GB713 Q345R                                   |                              | 0.6               |                |  |
| 10          | TK24324-03-02   | Cover lug 盖轴耳 I             | 1          | SA/GB713 Q345R                                   |                              | 0.7               |                |  |
| 9           | GB/T95-2002     | Plain Washer 平垫圈 24         | 2          | 100HV                                            | /                            | /                 |                |  |
| 8           | TK24324-03-01   | Shaft pin 轴销                | 1          | 20 (GB/T699)                                     |                              | 1.3               |                |  |
| 7           | ASME B18.22.1   | Plain Washer 平垫圈 NPS 1-1/8  | 16         | 100HV                                            | 0.05                         | 0.8               | Latest edition |  |
| 6           | ASME B18.2.2    | Nut 螺母 1-1/8-8UN-2B         | 16         | SA-194M-2H                                       | 0.09                         | 1.44              | Latest edition |  |
| 5           | ASME B18.2.1    | Bolt 螺栓 1-1/8-8UN-2A, L=135 | 16         | SA-193M-B7                                       | 0.18                         | 2.88              | Latest edition |  |
| 4           |                 | Gasket 垫片 0548×0458, T=3    | 1          | Elastomers with cotton fabric insertion 含有棉纤维的橡胶 |                              | /                 |                |  |
| 3           | ASME B16.5-2020 | Blind 法兰盖 BL 450-150 RF     | 1          | SA/GB713 Q345R                                   |                              | 98.5              |                |  |
| 2           | ASME B16.5-2020 | Flange 法兰 SO 450-150 RF     | 1          | SA-105M                                          |                              | 49.5              |                |  |
| 1           |                 | Pipe 接管 Ø457×12, L=233      | 1          | SA-516M Gr.485                                   |                              | 30.7              |                |  |
| 序号<br>No.   | 标准号<br>Standard | 名称<br>Name                  | 数量<br>QTY. | 材料<br>Material                                   | 单件<br>Single<br>重量<br>Weight | 总计<br>total<br>Kg | 备注<br>Remark   |  |
|             |                 |                             |            | Manhole<br>人孔<br>(DN450)                         |                              |                   |                |  |
|             |                 |                             |            |                                                  |                              |                   |                |  |
|             |                 |                             |            |                                                  |                              |                   |                |  |
| Rev.        | Comment         | Signature                   | Date       |                                                  |                              |                   |                |  |
| Design by   |                 | Date                        |            | 重量<br>Weight                                     | 188.4kg                      | 比例<br>Scale       | 1:10           |  |
| Approved by |                 | Date                        |            | Drawing NO.:                                     | TK24324-03                   | Rev. A            | Customer:      |  |

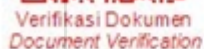
浙江泰克压力容器有限公司  
Zhejiang Tank Pressure Vessel Co., Ltd.

3. Material for manhole blind is SA/GB713 Q345R shall be normalized.  
人孔盖板材料SA/GB713 Q345R必须为正火板。

角焊缝的焊脚尺寸按较薄板的厚度。

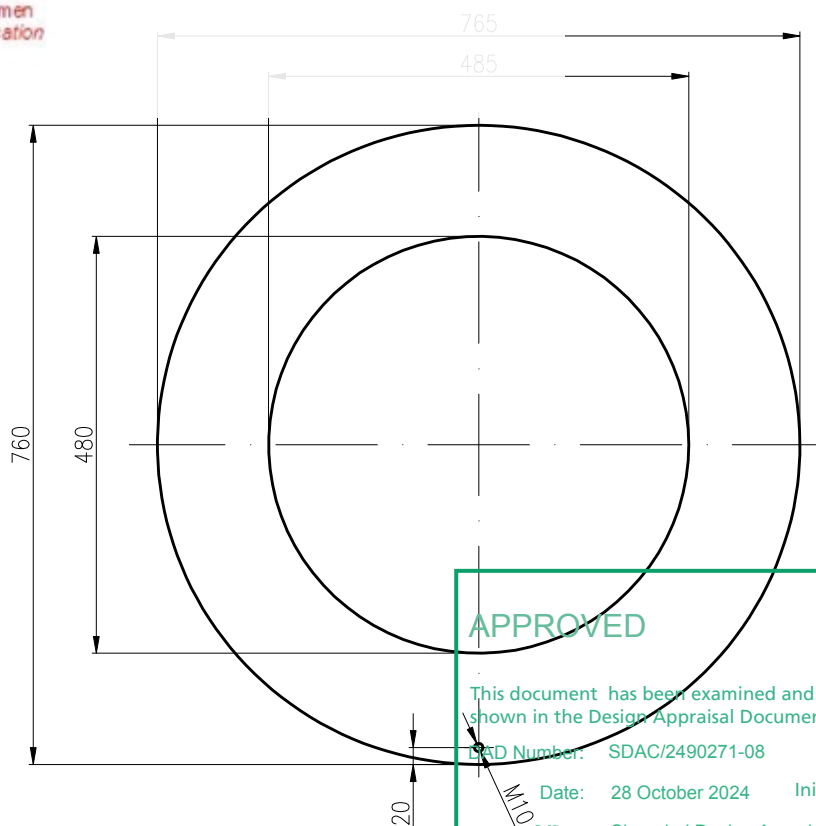
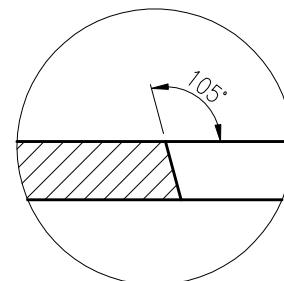
其余要求见装配图。





NO. VERIFIKASI: JKPP/2025/01-05/T-01/00542  
VERIFICATION NO.

TARIKH: 08/05/2025  
DATE



Developed LRQA Verif  
展开图

APPROVED

LRQA

This document has been examined and given the status as shown in the Design Appraisal Document (DAD) number below:

DAD Number: SDAC/2490271-08

Date: 28 October 2024      Initials: WL

Office: Shanghai Design Appraisal Cent

LROA Verification Limited

A member of the LRQA Group Limited

Notes:

1. Material材料: SA-516M Gr. 485.

2. Splicing is not allowed unless the provisions given below are satisfied.

补强圈允许拼接，但是要满足下列规定：

(1) Individual segments are joined by full penetration butt welds and shall be ground flush;  
各个分块部分都由全焊透对接焊缝连接, 并且焊缝磨平;

(2) Each segment of the reinforcing element shall have a vent hole as required by UG-37(g).  
每个分块都要按照UG-37(g)的要求具有一个信号孔。

(3) Each butt weld is radiographed or is oriented at least 45 deg from the longitudinal axis of the cylinder.

拼焊焊缝作射线检测或焊缝的布置由圆筒的纵轴起至少为 $45^\circ$ 。

|             |            |                 |         |                                      |        |             |     |                                                         |  |
|-------------|------------|-----------------|---------|--------------------------------------|--------|-------------|-----|---------------------------------------------------------|--|
|             |            |                 |         | Reinforcement pad<br>补强圈 DN450       |        |             |     | 浙江泰克压力容器有限公司<br>Zhejiang Tank Pressure Vessel Co., Ltd. |  |
|             |            |                 |         |                                      |        |             |     |                                                         |  |
| 版本<br>Rev.  | 注释 Comment | 签名<br>Signature | 日期 Date |                                      |        |             |     |                                                         |  |
| Design by   |            | Date            |         | 重量<br>Weight                         | 16.9kg | 比例<br>Scale | 1:1 | Customer:                                               |  |
| Approved by |            | Date            |         | Drawing NO.:<br>TK24324-04<br>Rev. A |        |             |     |                                                         |  |

JABATAN KESELAMATAN DAN KESIHATAN PEKERJAAN  
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



Verifikasi Dokumen  
Document Verification

NO. VERIFIKASI:  
VERIFICATION NO.

JKKP/2025/01-05/T-01/00542 China

TARIKH:  
DATE

08/05/2025

CODE: ASME SECT. VIII DIV. 1 2023 EDITION

EQUIPMENT NAME

Air receiver

VOLUME (V) (m<sup>3</sup>)

25

PRODUCT NO.

TK24324-XXX

WEIGHT (Kg)

4822

TEST DATE

XXXX/XX/XX

YEAR BUILT

XXXX

DESIGN PRESSURE (PS) (MPa)

0.84

MIN/MAX ALLOWABLE TEMPERATURE (TS) (°C)

-29/100

HYDRO. TEST PRESSURE (PT) (MPa)

1.1

RETAINED FOR INFORMATION

LRQA

As referred to in the Design Appraisal Document  
(DAD) number below:

DAD Number: SDAC/2490271-08

Date: 28 October 2024

Initials: WL

Office: Shanghai Design Appraisal Centre

NOTES:

1. Material材料::S30408 (GB/T24511-2017);

2. One piece for each vessel 每台1件

3. The required markings on a nameplate shall be in characters not less than  $\frac{5}{32}$  in. (4mm) high, and characters shall be either indented or raised at least 0.004 in. (0.10mm) and shall be legible and readable. 铭牌上所需的各种标志, 可采用高度不小于  $\frac{5}{32}$  in. (4mm) 的字符来标志, 标志字符应凹进或凸起至0.004in. (0.10mm), 并且明显易读。

|             |            |  |  |                 |         |  |  |                                                         |       |             |     |           |  |
|-------------|------------|--|--|-----------------|---------|--|--|---------------------------------------------------------|-------|-------------|-----|-----------|--|
|             |            |  |  | Nameplate<br>铭牌 |         |  |  | 浙江泰克压力容器有限公司<br>Zhejiang Tank Pressure Vessel Co., Ltd. |       |             |     |           |  |
|             |            |  |  |                 |         |  |  |                                                         |       |             |     |           |  |
|             |            |  |  |                 |         |  |  |                                                         |       |             |     |           |  |
| 版本<br>Rev.  | 注释 Comment |  |  | 签名<br>Signature | 日期 Date |  |  |                                                         |       |             |     |           |  |
| Design by   |            |  |  | Date            |         |  |  | 重量<br>Weight                                            | 0.5Kg | 比例<br>Scale | 1:1 |           |  |
| Approved by |            |  |  | Date            |         |  |  | Drawing NO.:<br>TK24324-05<br>Rev. A                    |       |             |     | Customer: |  |





JABATAN KESELAMATAN DAN KESIHATAN PEKERJAAN  
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH

NO. VERIFIKASI: JKKP/2025/01-05/T-01/00542  
VERIFICATION NO.

TARIKH: 08/05/2025  
DATE



Verifikasi Dokumen  
Document Verification

REMARKS:

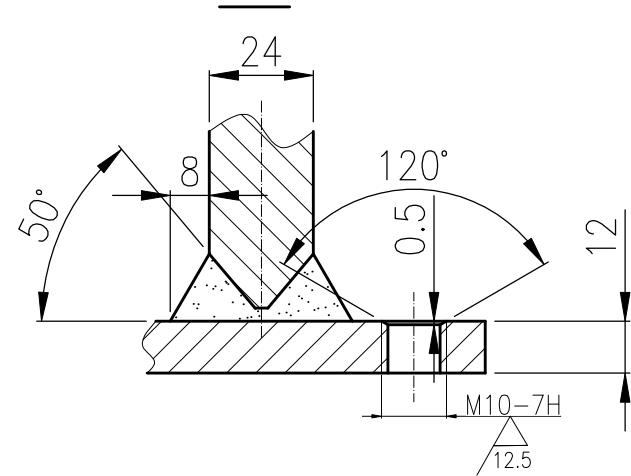
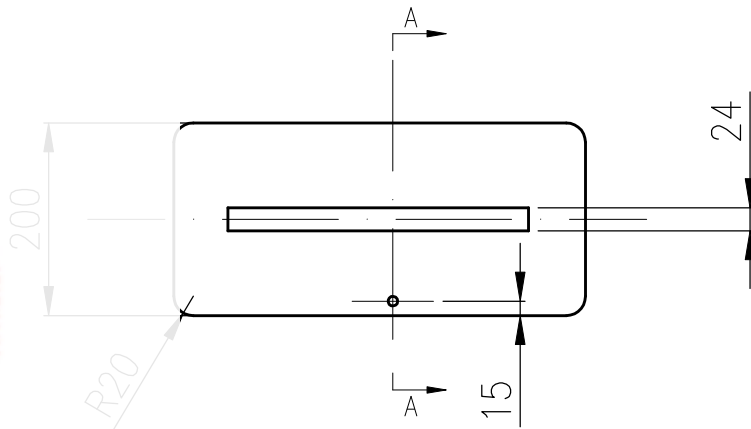
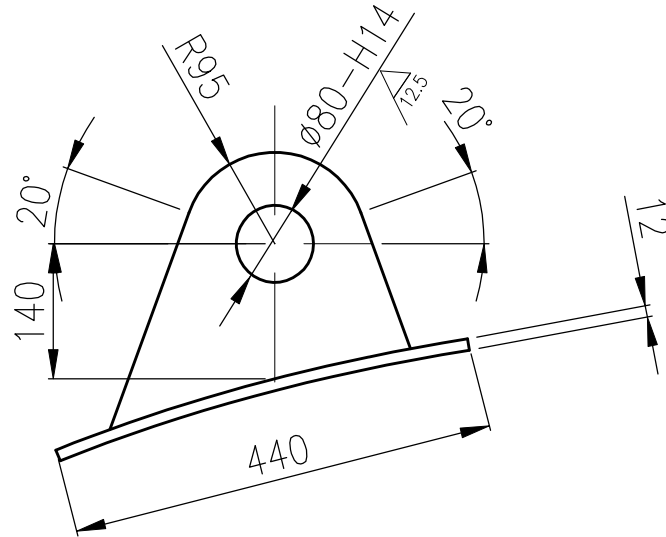
Lifting material: SA-516M Gr. 65. Lug material: Q235-B.

吊耳材料: Q235-B.

The lug shall fit the outer wall of the vessel.

During the welding of the lug, all parts shall be flat without

吊耳应平整，不得翘曲。



## RETAINED FOR INFORMATION

LRQA

As referred to in the Design Appraisal Document  
(DAD) number below:

DAD Number: SDAC/2490271-08

Date: 28 October 2024

Initials: WL

Office: Shanghai Design Appraisal Centre

LRQA Verification Limited

A member of the LRQA Group Limited

LRQA002 11.2021

|             |         |           |      |                                      |       |             |      |                                                         |  |
|-------------|---------|-----------|------|--------------------------------------|-------|-------------|------|---------------------------------------------------------|--|
|             |         |           |      | Lifting lug<br>吊耳                    |       |             |      | 浙江泰克压力容器有限公司<br>Zhejiang Tank Pressure Vessel Co., Ltd. |  |
| Rev.        | Comment | Signature | Date |                                      |       |             |      |                                                         |  |
| Design by   |         | Date      |      | 重量<br>Weight                         | 6.8kg | 比例<br>Scale | 1:10 |                                                         |  |
| Approved by |         | Date      |      | Drawing NO.:<br>TK24324-07<br>Rev. A |       |             |      | Customer:                                               |  |





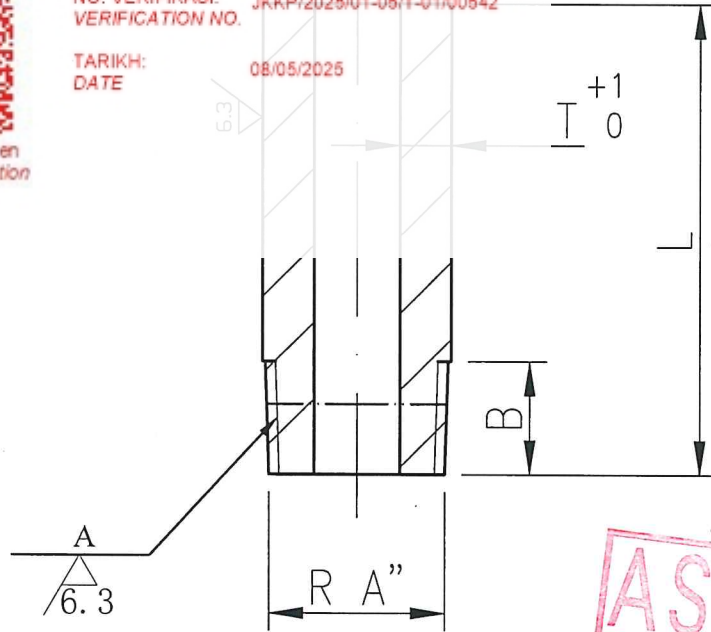
JABATAN KESELAMATAN DAN KESIHATAN PEKERJAAN  
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



Verifikasi Dokumen  
Document Verification

NO. VERIFIKASI:  
VERIFICATION NO. JKKP/2025/01-05/T-01/00542

TARIKH:  
DATE 08/05/2025



NOTES:

1. Material: See assembly dwg or purchase note.

材料: 见装配图明细表或采购订单。

2. Material marking should be in accordance with following requirements  
(a) Manufacturer's Name or Trademark; (b) Drawing number; (c) Size; (d) Material specification and designation; (e) Heat number;

材料标记需满足以下要求:

(a) 制造厂的名称或商标; (b) 图号; (c) 规格; (d) 材料标准号和牌号; (e) 炉号;  
3. Unless noted, otherwise processing surface tolerance shall be as per GB/T1804-2000 Grade m;

除注明外, 未注公差按GB/T1804-2000的m级;

4. End edges shall be chamfered or rounded.  
端部边缘应倒角或圆角。

5. Undeclared chamfer is 1X45°;  
未注倒角为1X45°。

others: 6.3/

APPROVED

LRQA

This document has been examined and given the status as shown in the Design Appraisal Document (DAD) number below:

DAD Number: SDAC/2490271-08

Date: 28 October 2024 Initials: WL

Office: Shanghai Design Appraisal Cent

LRQA Verification Limited

A member of the LRQA Group Limited

LRQA007 11.2021

|    |        |        |        |        |        |               |          |
|----|--------|--------|--------|--------|--------|---------------|----------|
| 16 | 3"     | 33.3   | 9.5    | 89     | 80     | 1.02          |          |
| 15 | 2-1/2" | 30.2   | 8      | 76     |        | 0.74          |          |
| 14 | 2"     | 25.7   | 9.15   | 60.3   |        | 0.58          |          |
| 13 | 1-1/2" | 21.4   | 8.15   | 48.3   |        | 0.40          |          |
| 12 | 1-1/4" | 21.4   | 6.1    | 42.2   |        | 0.24          |          |
| 11 | 1"     | 19.1   | 6.7    | 33.4   |        | 0.37          |          |
| 10 | 3/4"   | 16.3   | 6.35   | 26.7   |        | 0.27          |          |
| 09 | 1/2"   | 15     | 5.65   | 21.3   |        | 0.21          |          |
| 08 | 3"     | 33.3   | 9.5    | 89     | 50     | 0.93          |          |
| 07 | 2-1/2" | 30.2   | 8      | 76     |        | 0.67          |          |
| 06 | 2"     | 25.7   | 9.15   | 60.3   |        | 0.58          |          |
| 05 | 1-1/2" | 21.4   | 8.15   | 48.3   |        | 0.40          |          |
| 04 | 1-1/4" | 21.4   | 6.1    | 42.2   |        | 0.27          |          |
| 03 | 1"     | 19.1   | 6.7    | 33.4   |        | 0.23          |          |
| 02 | 3/4"   | 16.3   | 6.35   | 26.7   |        | 0.17          |          |
| 01 | 1/2"   | 15     | 5.65   | 21.3   |        | 0.13          |          |
| XX | A      | B (mm) | T (mm) | D (mm) | L (mm) | 重量weight (kg) | 备注Remark |

|             |         |  |           |                                 |                                     |   |                                                         |           |
|-------------|---------|--|-----------|---------------------------------|-------------------------------------|---|---------------------------------------------------------|-----------|
|             |         |  |           | Screwed joint (R)<br>插入式Rp外螺纹接头 |                                     |   | 浙江泰克压力容器有限公司<br>Zhejiang Tank Pressure Vessel Co., Ltd. |           |
| Rev.        | Comment |  | Signature | Date                            |                                     |   |                                                         |           |
| Design by   | 田家欢     |  | Date      | 2023.11.7                       | 重量<br>Weight                        | / | 比例<br>Scale                                             | /         |
| Approved by | 田家欢     |  | Date      | 2023.11.7                       | Drawing NO.:<br>TKTY02-XX<br>Rev. C |   |                                                         | Customer: |